

its rays starting at $\emptyset$, and hence to the infinite 0–1 sequences.

These examples suggest that the ends of a graph can be thought of as ‘points at infinity’ to which its rays converge. We shall formalize this in Section 8.6, where we define a natural topology on a graph and its ends in which rays will indeed converge to their respective ends.

The maximum number of disjoint rays in an end is the (*combinatorial*) *vertex-degree* of that end, the maximum number of edge-disjoint rays in it is its (*combinatorial*) *edge-degree*. These maxima are indeed attained: if an end contains a set of k (edge-) disjoint rays for every integer k , it also contains an infinite set of (edge-) disjoint rays (Exercise 46). Thus, every end has a vertex-degree and an edge-degree in $\mathbb{N} \cup \{\infty\}$. end degrees

8.2 Paths, trees, and ends

There are two fundamentally different aspects to the infinity of an infinite connected graph: one of ‘length’, expressed in the presence of rays, and one of ‘width’, expressed locally by infinite degrees. The infinity lemma tells us that at least one of these must occur:

Proposition 8.2.1. *Every infinite connected graph has a vertex of infinite degree or contains a ray.*

Proof. Let G be an infinite connected graph with all degrees finite. Let v_0 be a vertex, and for every $n \in \mathbb{N}$ let V_n be the set of vertices at distance n from v_0 . Induction on n shows that the sets V_n are finite, and hence that $V_{n+1} \neq \emptyset$ (because G is infinite and connected). Furthermore, the neighbour of a vertex $v \in V_{n+1}$ on any shortest $v-v_0$ path lies in V_n . By Lemma 8.1.2, G contains a ray. (8.1.2) $\square$

Often it is useful to have more detailed information on how this ray or vertex of infinite degree lies in G . The following lemma enables us to find it ‘close to’ any given infinite set of vertices.

Lemma 8.2.2. (Star-Comb Lemma)

Let U be an infinite set of vertices in a connected graph G . Then G contains either a comb with all teeth in U or a subdivision of an infinite star with all leaves in U . [8.6.3]

Proof. As G is connected, it contains a path between two vertices in U . This path is a tree $T \subseteq G$ every edge of which lies on a path in T between two vertices in U . By Zorn’s lemma there is a maximal such tree T^* . Since U is infinite and G is connected, T^* is infinite. If T^* has a vertex of infinite degree, it contains the desired subdivided star.

Suppose now that T^* is locally finite. Then T^* contains a ray R (Proposition 8.2.1). Let us construct a sequence $P_1, P_2, \dots$ of disjoint R - U paths in T^* . Having chosen P_i for every $i < n$ for some n , pick $v \in R$ so that vR meets none of those paths P_i . The first edge of vR lies on a path P in T^* between two vertices in U ; let us think of P as traversing this edge in the same direction as R , and choose P minimal. Then vP has the form $vRwP$, where $P_n := wP$ is an R - U path. And $P_n \cap P_i = \emptyset$ for all $i < n$, because $P_i \cup Rv \cup P_n$ contains no cycle. $\square$

We shall often apply Lemma 8.2.2 in locally finite graphs, in which case it always yields a comb.

Recall that a rooted tree $T \subseteq G$ is *normal* in G if the endvertices of every T -path in G are comparable in the tree-order of T . If T is a spanning tree, the only T -paths are edges of G that are not edges of T .

Normal spanning trees are perhaps the single most important structural tool in infinite graph theory. As in finite graphs, they exhibit the separation properties of the graph they span.⁶ Moreover, their *normal rays*, those that start at the root, reflect its end structure:

normal ray

[8.6.8] **Lemma 8.2.3.** *If T is a normal spanning tree of G , then every end of G contains exactly one normal ray of T .*

(1.5.4) *Proof.* Let $\omega \in \Omega(G)$ be given. Apply the star-comb lemma in T with U the vertex set of any ray $R \in \omega$. If the lemma gives a subdivided star with leaves in U and centre z , say, then the finite down-closure $[z]$ of z in T separates infinitely many vertices $u > z$ of U pairwise in G (Lemma 1.5.4). This contradicts our choice of U .

So T contains a comb with teeth on R . Let $R' \subseteq T$ be its spine. Since every ray in T has an increasing tail (Exercise 4), we may assume that R' is a normal ray. Since R' is equivalent to R , it lies in ω .

Conversely, distinct normal rays of T are separated in G by the (finite) down-closure of their greatest common vertex (Lemma 1.5.4), so they cannot belong to the same end of G . $\square$

Not all connected graphs have a normal spanning tree; complete uncountable graphs, for example, have none. (Why not?) The quest to characterize the graphs that have a normal spanning tree is not entirely over, and it has held some surprises.⁷ One of the most useful sufficient conditions is that the graph contains no $TK^{\aleph_0}$; see Theorem 12.6.9. For our purposes, the following result suffices:

[8.7.2]

Theorem 8.2.4. (Jung 1967)

Every countable connected graph has a normal spanning tree.

⁶ Lemma 1.5.4 continues to hold for infinite graphs, with the same proof.

⁷ One of these is Theorem 8.6.2; for more see the notes.

Proof. The proof follows that of Proposition 1.5.5; we only sketch the differences. Starting with a single vertex, we construct an infinite sequence $T_0 \subseteq T_1 \subseteq \dots$ of finite normal trees in G , all with the same root, whose union T will be a normal spanning tree. (1.5.5)

To ensure that T spans G , we fix an enumeration $v_0, v_1, \dots$ of $V(G)$ and see to it that T_n contains v_n . It is clear that T will be a tree (since any cycle in T would lie in some T_n , and every two vertices of T lie in a common T_n and can be linked there), and clearly the tree order of T induces that of the T_n . Finally, T will be normal, because the endvertices of any edge of G that is not an edge of T lie in some T_n : since that T_n is normal, they must be comparable there, and hence in T .

It remains to specify how to construct T_{n+1} from T_n . If $v_{n+1} \in T_n$, put $T_{n+1} := T_n$. If not, let C be the component of $G - T_n$ containing v_{n+1} . Let x be the greatest element of the chain $N(C)$ in T_n , and let T_{n+1} be the union of T_n and an x - v_{n+1} path P with $\dot{P} \subseteq C$. Then the neighbourhood in T_{n+1} of any new component $C' \subseteq C$ of $G - T_{n+1}$ is a chain in T_{n+1} , so T_{n+1} is again normal. $\square$

One of the most basic problems in an infinite setting that has no finite equivalent is whether or not ‘arbitrarily many’, in some context, implies ‘infinitely many’. Suppose we can find k disjoint rays in some given graph G , for every $k \in \mathbb{N}$; does G also contain an infinite set of disjoint rays?

The answer to the corresponding question for finite paths (of any fixed length) is clearly ‘yes’, since a finite path P can never get in the way of more than $|P|$ disjoint other paths. A badly chosen ray, however, can meet infinitely many other rays, preventing them from being selected for the same disjoint set. Rather than collecting our disjoint rays greedily, we therefore have to construct them carefully and all simultaneously.

The proof of the following theorem is a nice example of a construction in an infinite sequence of steps, where the final object emerges only at the limit step. Each of the steps in the sequence will involve a non-trivial application of Menger’s theorem (3.3.1).

Theorem 8.2.5. (Halin 1965)

- (i) If an infinite graph G contains k disjoint rays for every $k \in \mathbb{N}$, then G contains infinitely many disjoint rays.
- (ii) If an infinite graph G contains k edge-disjoint rays for every $k \in \mathbb{N}$, then G contains infinitely many edge-disjoint rays.

Proof. (i) We construct our infinite system of disjoint rays inductively in ω steps. After step n , we shall have found n disjoint rays $R_1^n, \dots, R_n^n$ and chosen initial segments $R_i^n x_i^n$ of these rays. In step $n+1$ we choose the rays $R_1^{n+1}, \dots, R_{n+1}^{n+1}$ so as to extend these initial segments, i.e. so that $R_i^n x_i^n$ is a proper initial segment of $R_i^{n+1} x_i^{n+1}$, for $i = 1, \dots, n$. (3.3.1)

Then, clearly, the graphs $R_i^* := \bigcup_{n \in \mathbb{N}} R_i^n x_i^n$ will form an infinite family $(R_i^*)_{i \in \mathbb{N}}$ of disjoint rays in G .

n
 R_i, x_i For $n = 0$ the empty set of rays is as required. So let us assume that $R_1^n, \dots, R_n^n$ have been chosen, and describe step $n + 1$. For simplicity, let us abbreviate $R_i^n =: R_i$ and $x_i^n =: x_i$. Let $\mathcal{R}$ be any set of $|R_1 x_1 \cup \dots \cup R_n x_n| + n^2 + 1$ disjoint rays (which exists by assumption), and immediately delete those rays from $\mathcal{R}$ that meet any of the paths $R_1 x_1, \dots, R_n x_n$; then $\mathcal{R}$ still contains at least $n^2 + 1$ rays.

I
 m
 Z We begin by repeating the following step as often as possible. If there exists an $i \in \{1, \dots, n\}$ such that R_i^{n+1} has not yet been defined and $\dot{x}_i R_i$ meets at most n of the rays currently in $\mathcal{R}$, we delete those rays from $\mathcal{R}$, put $R_i^{n+1} := R_i$, and choose as x_i^{n+1} the successor of x_i on R_i . Having performed this step as often as possible, we let I denote the set of those $i \in \{1, \dots, n\}$ for which R_i^{n+1} is still undefined, and put $|I| =: m$. Then $\mathcal{R}$ still contains at least $n^2 + 1 - (n - m)n \geq m^2 + 1$ rays. Every R_i with $i \in I$ meets more than $n \geq m$ of the rays in $\mathcal{R}$; let z_i be its first vertex on the m th ray it meets. Then $Z := \bigcup_{i \in I} x_i R_i z_i$ meets at most m^2 of the rays in $\mathcal{R}$; we delete all the other rays from $\mathcal{R}$, choosing one of them as R_{n+1}^{n+1} (with x_{n+1}^{n+1} arbitrary).

y_i On each remaining ray $R \in \mathcal{R}$ we now pick a vertex $y = y(R)$ after its last vertex in Z , and put $Y := \{y(R) \mid R \in \mathcal{R}\}$. Let H be the union of Z and all the paths Ry ($R \in \mathcal{R}$). Then $X := \{x_i \mid i \in I\}$ cannot be separated from Y in H by fewer than m vertices, because these would miss both one of the m rays R_i with $i \in I$ and one of the m rays in $\mathcal{R}$ that meet $x_i R_i z_i$ for this i . So by Menger's theorem (3.3.1) there are m disjoint X - Y paths $P_i = x_i \dots y_i$ ($i \in I$) in H . For each $i \in I$ let R'_i denote the ray from $\mathcal{R}$ that contains y_i , choose as R_i^{n+1} the ray $R_i x_i P_i y_i R'_i$, and put $x_i^{n+1} := y_i$.

(ii) The proof is similar; see Exercise 45 and its hint. $\square$

nH Does Theorem 8.2.5 generalize to other graphs than rays? Let us call a graph H *ubiquitous* with respect to a relation $\leq$ between graphs (such as the subgraph relation $\subseteq$, or the minor relation $\preceq$) if $nH \leq G$ for all $n \in \mathbb{N}$ implies $\aleph_0 H \leq G$, where nH denotes the disjoint union of n copies of H . Ubiquity appears to be closely related to questions of well-quasi-ordering as discussed in Chapter 12. Non-ubiquitous graphs exist for all the standard graph orderings; see Exercise 49 for an example of a locally finite graph that is not ubiquitous under the subgraph relation.

Ubiquity conjecture. (Andreae 2002)

Every locally finite connected graph is ubiquitous with respect to the minor relation.

A proof of the ubiquity conjecture for trees is indicated in Exercise 50.

Just as in Theorem 8.2.5 one can show that an end contains infinitely many disjoint rays as soon as the number of disjoint rays in it is not finitely bounded, and similarly for edge-disjoint rays (Exercise 46). Hence, the maxima in our earlier definitions of the vertex- and edge-degrees of an end exist as claimed. Ends of infinite vertex-degree are called *thick*; ends of finite vertex-degree are *thin*.

thick/thin

The $\mathbb{N} \times \mathbb{N}$ *grid*, for example, the graph on $\mathbb{N}^2$ in which two vertices (n, m) and (n', m') are adjacent if and only if $|n - n'| + |m - m'| = 1$, has only one end, which is thick. In fact, the $\mathbb{N} \times \mathbb{N}$ grid is a kind of prototype for thick ends: every graph with a thick end contains it as a minor. This is another classical result of Halin, which we prove in the remainder of this section.

grid

For technical reasons, we shall prove Halin's theorem for hexagonal rather than square grids. These may seem a little unwieldy at first, but have the advantage that they can be found as topological rather than ordinary minors (Proposition 1.7.3), which makes them much easier to handle. We shall define the hexagonal grid H^∞ so that it is a subgraph of the $\mathbb{N} \times \mathbb{N}$ grid, and it will be easy to see that, conversely, the $\mathbb{N} \times \mathbb{N}$ grid is a minor of H^∞ . (See also Exercise 71, Ch. 12.)

To define our standard copy of the *hexagonal quarter grid* H^∞ , we delete from the $\mathbb{N} \times \mathbb{N}$ grid H the vertex $(0, 0)$, the vertices (n, m) with $n > m$, and all edges $(n, m)(n + 1, m)$ such that n and m have equal parity (Fig. 8.2.1). Thus, H^∞ consists of the *vertical rays*

 H^∞

$$\begin{aligned} U_0 &:= H[\{(0, m) \mid 1 \leq m\}] \\ U_n &:= H[\{(n, m) \mid n \leq m\}] \quad (n \geq 1) \end{aligned}$$

 U_n

and between these a set of *horizontal edges*,

$$E := \{(n, m)(n + 1, m) \mid n \not\equiv m \pmod{2}\}.$$

To enumerate these edges, as $e_1, e_2, \dots$ say, we order them colexicographically: the edge $(n, m)(n + 1, m)$ precedes the edge $(n', m')(n' + 1, m')$ if $m < m'$, or if $m = m'$ and $n < n'$ (Fig. 8.2.1).

 $e_1, e_2, \dots$

Theorem 8.2.6. (Halin 1965)

Whenever a graph contains a thick end, it has a TH^∞ subgraph whose rays belong to that end.

Proof. Given two infinite sets $\mathcal{P}, \mathcal{P}'$ of finite or infinite paths, let us write $\mathcal{P} \geq \mathcal{P}'$ if $\mathcal{P}'$ consists of final segments of paths in $\mathcal{P}$. (Thus, if $\mathcal{P}$ is a set of rays, then so is $\mathcal{P}'$.)

(8.1.2)
 $\leq$

Let G be any graph with a thick end ω . Our task is to find disjoint rays in ω that can serve as ‘vertical’ (subdivided) rays U_n for our desired grid, and to link these up by suitable disjoint ‘horizontal’ paths. We

 ω

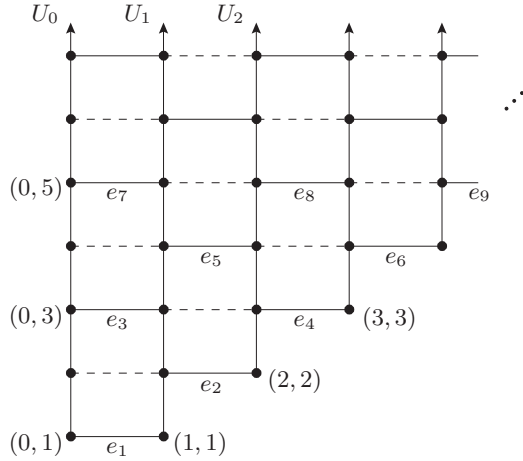


Fig. 8.2.1. The hexagonal quarter grid H^∞ .

begin by constructing a sequence $R_0, R_1, \dots$ of rays (of which we shall later choose some tails R'_n as ‘vertical rays’), together with path systems $\mathcal{P}_n$ between the R_n and suitable $R_{p(n)}$ with $p(n) < n$ (from which we shall later choose the ‘horizontal paths’). We shall aim to find the R_n in ‘supply sets’ $\mathcal{R}_0 \geq \mathcal{R}_1 \geq \dots$ of unused rays. After the n th construction step we shall have constructed the subgraph $H_n := \bigcup_{i=0}^n (R_i \cup \mathcal{P}_i)$.

We start with any infinite set $\mathcal{R}$ of disjoint rays in ω ; this exists by our assumption that ω is a thick end. Pick $R_0 \in \mathcal{R}$, put $\mathcal{P}_0 := \emptyset$, and let $\mathcal{R}_0 := \mathcal{R} \setminus \{R_0\}$. At step $n \geq 1$ of the construction we shall choose the following:

- (1) a ray $R_n \in \omega$ disjoint from H_{n-1} ;
- (2) an integer $p(n) < n$;
- (3) an infinite set $\mathcal{P}_n$ of disjoint $R_n - R_{p(n)}$ paths avoiding every other R_i ;
- (4) an infinite set $\mathcal{R}_n \leq \mathcal{R}_{n-1}$ of disjoint rays in $G - H_n$.

Let $n \geq 1$ be given. As a first candidate for R_n consider any ray $R \in \mathcal{R}_{n-1}$. By (4) for smaller values of n , we have both $R \in \omega$ (since $\mathcal{R}_{n-1} \leq \dots \leq \mathcal{R}_0 \subseteq \mathcal{R}$) and $R \cap H_{n-1} = \emptyset$, as required for R_n in (1).

Next, let us try to find a set $\mathcal{P}_n$ and $p(n)$ to go with R as R_n . Since H_{n-1} contains $R_0 \in \omega$, there exists an infinite set $\mathcal{P}$ of disjoint $R - H_{n-1}$ paths in G . If $\mathcal{P}$ has an infinite subset of paths all ending on the same R_i ($i < n$), we delete all other paths from $\mathcal{P}$. If not, then $\mathcal{P}$ has an infinite subset of paths all ending at an inner vertex of a path in $\mathcal{P}_i$, for the same $i < n$. We extend them back along this path of $\mathcal{P}_i$ until they hit R_i , and delete all other paths from $\mathcal{P}$. In both cases we

put $p(n) := i$, satisfying (2), and have found an infinite set $\mathcal{P}$ of disjoint R - $R_{p(n)}$ paths that avoid all the other R_i ($i < n$).

For later use we record:

Every path in $\mathcal{P}$ consists of an R - H_{n-1} path followed by a (possibly trivial) path in H_{n-1} . (*)

What can still prevent us from choosing R as R_n and $\mathcal{P}$ as $\mathcal{P}_n$ is condition (4): if $\mathcal{P}$ meets all but finitely many rays in $\mathcal{R}_{n-1}$ infinitely, we cannot find an infinite set $\mathcal{R}_n \leq \mathcal{R}_{n-1}$ of rays avoiding $\mathcal{P}$.

If this happens, the only option we have is to make a virtue of necessity and use the rich intersection of $\mathcal{P}$ and $\mathcal{R}_{n-1}$ for an altogether different construction of R_n , $\mathcal{P}_n$ and $\mathcal{R}_n$. This will require some work. But we may assume the following, as the result of our effort so far:

Whenever $R' \in \mathcal{R}_{n-1}$ and $\mathcal{P}' \leq \mathcal{P}$ is an infinite set of R' - $R_{p(n)}$ paths, there is a ray $R'' \neq R'$ in $\mathcal{R}_{n-1}$ that meets $\mathcal{P}'$ infinitely. (**)

For if (**) failed, we could choose R' as R_n and $\mathcal{P}'$ as $\mathcal{P}_n$, and for $\mathcal{R}_n$ select from every ray $R'' \neq R'$ in $\mathcal{R}_{n-1}$ a tail avoiding $\mathcal{P}'$. This would satisfy conditions (1)–(4) for n .

Consider the paths in $\mathcal{P}$ as linearly ordered by the natural order of their starting vertices on R . This induces an ordering on every $\mathcal{P}' \leq \mathcal{P}$. If $\mathcal{P}'$ is a set of R' - $R_{p(n)}$ paths for some ray R' , we shall call this ordering of $\mathcal{P}'$ *compatible* with R' if the ordering it induces on the first vertices of its paths coincides with the natural ordering of those vertices on R' .

Starting with $R =: R_{n-1}^0$ and $\mathcal{P} =: \mathcal{P}^0$, let us construct sequences $R_{n-1}^0, R_{n-1}^1, \dots$ and $\mathcal{P}^0 \geq \mathcal{P}^1 \geq \dots$ such that every R_{n-1}^k is a tail of a ray in $\mathcal{R}_{n-1}$ and each $\mathcal{P}^k$ is an infinite set of R_{n-1}^k - $R_{p(n)}$ paths whose ordering is compatible with R_{n-1}^k . The first path of $\mathcal{P}^k$ in this ordering will be denoted by P_k , its starting vertex on R_{n-1}^k by v_k , and the path in $\mathcal{P}^{k-1}$ containing P_k (if $k \geq 1$) by P_k^- (Fig. 8.2.2). To define R_{n-1}^k and $\mathcal{P}^k$ for $k \geq 1$, we use (**) with $R' \supseteq R_{n-1}^{k-1}$ and $\mathcal{P}' = \mathcal{P}^{k-1}$ to find in $\mathcal{R}_{n-1}$ a ray $R'' \not\supseteq R_{n-1}^{k-1}$ that meets $\mathcal{P}^{k-1}$ infinitely; let R_{n-1}^k be a tail of R'' that avoids the finitely many paths in $\mathcal{P}$ containing $P_0, \dots, P_{k-1}$. Let P_k^- be a path in $\mathcal{P}^{k-1}$ that meets R_{n-1}^k and let v be its ‘highest’ vertex on R_{n-1}^k , that is, the last vertex of R_{n-1}^k in $V(P_k^-)$. Replacing R_{n-1}^k with its tail vR_{n-1}^k , we can arrange that P_k^- has only the vertex v on R_{n-1}^k . Then $P_k := vP_k^-$ is an R_{n-1}^k - $R_{p(n)}$ path starting at $v_k = v$. We may now select an infinite set $\mathcal{P}^k \leq \mathcal{P}^{k-1}$ of R_{n-1}^k - $R_{p(n)}$ paths compatible with R_{n-1}^k and containing P_k as its first path.

Note that P_k cannot be a subpath of any P_i with $i < k$, since P_k contains $v_k \in R_{n-1}^k$ but $R_{n-1}^k \cap P_i = \emptyset$. As $\mathcal{P}^k \leq \mathcal{P}^i$, this means that P_k and P_i are subpaths of disjoint paths in $\mathcal{P}$. Similarly, for $i < k$ the rays R_{n-1}^k and R_{n-1}^i cannot be tails of the same ray in $\mathcal{R}_{n-1}$, and are there-

$p(n)$, $\mathcal{P}$

$R_{n-1}^0, \mathcal{P}^0$

P_k, v_k
 P_k^-

R_{n-1}^k

$\mathcal{P}^k$

It remains to verify (1). We have $R_n \in \omega$, since $\mathcal{P}_n$ joins R_n disjointly to $R_{p(n)}$, and $R_{p(n)} \in \omega$ by (1). To show that $R_n \cap H_{n-1} = \emptyset$, recall that the ‘vertical’ segments of R_n lie in rays from $\mathcal{R}_{n-1}$, so by (4) they do not meet H_{n-1} . And a ‘horizontal’ segment of R_n can meet H_{n-1} only if it does so also in its last vertex v_k , by (*). But v_k also lies on a vertical segment, and hence not in H_{n-1} . Thus, $R_n \cap H_{n-1} = \emptyset$ as desired.

Let us now use our rays R_n and path systems $\mathcal{P}_n$ to construct the desired grid. In the tree on $\mathbb{N}$ defined by joining each n to $p(n)$, apply the infinity lemma (8.1.2) to the distance classes from 0 to find a ray $n_0 n_1 \dots$ with vertices $n_0 < n_1 < \dots$ or, if one of these classes is infinite, a vertex n_0 with infinitely many neighbours $n_1, n_2, \dots$ greater than n_0 . We treat these two cases in turn, assuming for notational simplicity that $n_i = i$ for all i . (In other words, we discard any R_n with $n \notin \{n_0, n_1, \dots\}$.)

In the first case, each $\mathcal{P}_n$ is an infinite set of disjoint $R_n - R_{n-1}$ paths. Our aim is to choose tails R'_n of our rays R_n that will correspond to the vertical rays $U_n \subseteq H^\infty$, and paths $S_1, S_2, \dots$ between the R'_n that will correspond to the horizontal edges $e_1, e_2, \dots$ of H^∞ . We shall find the paths $S_1, S_2, \dots$ inductively, choosing the R'_n as needed as we go along (but also in the order of increasing n , starting with $R'_0 := R_0$). At every step of the construction, we shall have selected only finitely many S_k and only finitely many R'_n .

Let k and n be minimal such that S_k and R'_n are still undefined. We describe how to choose S_k , and R'_n if the definition of S_k requires it. Let i be such that e_k joins U_{i-1} to U_i in H^∞ . If $i = n$, let R'_n be a tail of R_n that avoids the finitely many paths $S_1, \dots, S_{k-1}$; otherwise, R'_i has already been defined, and so has R'_{i-1} . Now choose $S_k \in \mathcal{P}_i$ ‘high enough’ between R'_{i-1} and R'_i to mirror the position of e_k in H^∞ , and to avoid $S_1 \cup \dots \cup S_{k-1}$. Then S_k will also avoid every other R'_j already defined: by (3) for i if $j < i$, and by (1) for j if $j > i$. Since every R'_n is chosen so as to avoid all previously defined S_k , and every S_k avoids all previously defined R'_j (except R'_{i-1} and R'_i), the R'_n and S_k are pairwise disjoint for all $n, k \in \mathbb{N}$, except for the required incidences. Our construction thus yields the desired subdivision of H^∞ .

In the second case, every $\mathcal{P}_n$ is a set of disjoint $R_n - R_0$ paths. We now use only the R_n with $n \geq 1$ for vertical rays of H^∞ , because R_0 will be needed for the horizontal paths. More precisely, we choose rays $R'_n \subseteq R_n$ for $n \geq 1$, and paths S_k between them, inductively as before, except that S_k now consists of three parts: an initial segment from $\mathcal{P}_{i-1}$, followed by a middle segment on R_0 , and a final segment from $\mathcal{P}_i$. Such S_k can again be found, since at every stage of the construction only a finite part of R_0 has been used. $\square$